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The Width Tax

Electrical, Package, and Thermal Limits of Bigger PCIe Links

Amrendra Singh and DeepSeek V4.1 Flash 1017

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Abstract

PCI Express scales bandwidth by adding lanes. The x16 connector has been the top of the CEM standard for two decades, and the obvious next step—x32—is blocked less by the protocol than by physics: pins, package escapes, crosstalk, SerDes power, and thermal density. This paper builds a first-order lane budget that treats width, rate, modulation, and reach as a single optimization. We show that each additional lane carries a superlinear width tax in connector density, package layers, and local thermal load; that the alternative—fewer, faster lanes—trades pins for channel loss and equalization power; that lane merging and width reduction are useful for yield, power, and latency-bound workloads but not for peak bandwidth; and that no lanes at all—optics, wireless, and package-level fabrics—wins at the extremes of reach. The result is a lane frontier: parallel buses at short reach, PCIe lanes in the middle, and optics beyond a few meters. The next PCIe generation should expose width, rate, and modulation as a tunable triad rather than treating lane count as the only scaling knob.

Keywords: PCI Express; lanes; SerDes; PAM4; interconnect; thermal; packaging.

1. Introduction

PCI Express has one scaling knob that system builders trust: add lanes. The x16 link has been the top of the CEM connector for two decades, and the bandwidth curve has been delivered by raising the per-lane rate, not by widening the link [1, 2, 4]. The next obvious step is x32. The protocol already permits it; the connector, the package, and the channel do not.

This paper argues that the lane is a middle-distance compromise. At short reach, a wide parallel bus is cheaper per bit; at long reach, optics is cheaper per meter; in between, PCIe's lane count is a reasonable but not optimal choice. Adding lanes at the top end buys bandwidth linearly and pays a superlinear width tax in pins, package escapes, crosstalk, power, and thermal density. The alternatives—fewer faster lanes, wider modulation, lane merging, and lane-less fabrics—each trade one tax for another.

Our contributions are five. First, we define a lane budget that treats width, rate, modulation, and reach as one optimization (Section 2). Second, we quantify the width tax at x16, x32, and x64 (Section 3). Third, we analyze the three routes that avoid more lanes: fewer but faster lanes, wider modulation, and wider parallel lanes (Section 4). Fourth, we analyze lane merging and width reduction, including their effect on latency-bound memory traffic (Section 5). Fifth, we analyze the lane-less alternatives—optics, wireless, package fabrics, and network fabrics—and combine everything into a lane frontier (Sections 6 and 7).

The method is analytical and first-order. We take rates, widths, and overheads from public specifications and published architecture descriptions, and we build a model rather than a calibrated simulator. The parameters are listed in the tables so that a reader can substitute their own. We do not report measurements, and we flag estimates as estimates.

2. The Lane Budget

2.1 Definitions

A lane is one differential pair carrying one serial stream. A link of width N has $2N$ data wires plus sidebands and grounds. The lane rate R is measured in gigabaud; the modulation W is the number of bits per symbol (1 for NRZ, 2 for PAM4, 3 for PAM8); the efficiency η accounts for encoding, framing, FEC, and protocol overhead. The link bandwidth is

$$B = NRW\eta. \tag{1}$$

Table 1: PCIe generations. Bandwidth is one direction after encoding and framing; reach is a typical PCB limit without retimers and is an estimate.

Gen	Rate (GT/s)	Modulation	x16 (GB/s)	Per lane (GB/s)	Typical PCB reach
1.0	2.5	NRZ, 8b/10b	4	0.25	~1 m
2.0	5	NRZ, 8b/10b	8	0.50	~0.5 m
3.0	8	NRZ, 128b/130b	15.75	0.98	~0.3 m
4.0	16	NRZ, 128b/130b	31.5	1.97	~0.2 m
5.0	32	NRZ, 128b/130b	63	3.94	~0.15 m
6.0	64	PAM4, FLIT/FEC	121	7.56	~0.1 m
7.0	128	PAM4, FLIT/FEC	242	15.1	~0.05 m

Every route in this paper changes one of N, R, W, or η.

2.2 Cost

Cost is the sum of four terms:

$$P = Np(R, W) + p_{\text{ret}}(L) + p_{\text{conn}}, \tag{2}$$

$$A = Na(R, W) + a_{\text{conn}}, \tag{3}$$

$$T = PR_{\text{th}}, \tag{4}$$

where p is the per-lane SerDes power, p_{ret} the retimer power (zero below the reach limit), a the per-lane package and connector area, and R_{th} the thermal resistance. The width tax is the derivative $\partial P/\partial N$: it is not constant, because crosstalk, equalization, and retimers grow with N.

2.3 Reach

Reach is set by the channel loss and the SNR budget:

$$L_{\max} = \frac{S - S_0}{\alpha(R, W)}, \tag{5}$$

where S is the available SNR, S_0 the required SNR, and α the loss per meter. α grows with frequency and with the modulation order: PAM4 needs about 9.5 dB more SNR than NRZ at the same symbol rate, and PAM8 about 19 dB more [9, 10].

3. More Lanes: The Width Tax

3.1 The connector

The CEM connector has 164 pins. An x16 link uses 64 data pins (32 differential pairs), plus sidebands, grounds, and power. An x32 link needs 128 data pins; an x64 link needs 256. The connector is the first wall: a 328-pin or 656-pin edge connector is not a consumer part. Mating force, wipe, coplanarity, and cost all grow faster than the pin count. The connector is also where the channel is worst: the edge fingers, the vias, and the card-edge launch add reflections that the package and cable do not [3].

3.2 The package

Each lane needs two bumps, a ground return, and an escape route. At a 0.8–1 mm bump pitch, 32 lanes consume 64 bumps plus grounds; 64 lanes consume 128. The escape routing consumes substrate layers, and the layers consume cost and yield. The package edge is the second wall: the die's beachfront is fixed, and the SerDes area grows with N.

3.3 Electricity

More lanes mean more aggressors. Near-end and far-end crosstalk grow with the number of adjacent pairs; ground pins must be interleaved; simultaneous switching noise grows with the number of drivers switching at once. The differential pair is not immune: the pair's own skew and the pair-to-pair coupling set the eye. At x32, the channel is a 64-pair bundle, and the bundle is a shared noise environment [11].

3.4 Power

A 32 GT/s SerDes consumes roughly 50–100 mW per lane in 7 nm; 64 GT/s PAM4 roughly 100–150 mW. An x16 Gen6 link is 1.6–2.4 W; x32 is 3.2–4.8 W; x64 is 6.4–9.6 W. Add retimers at 1–2 W each: x32 needs two to four, x64

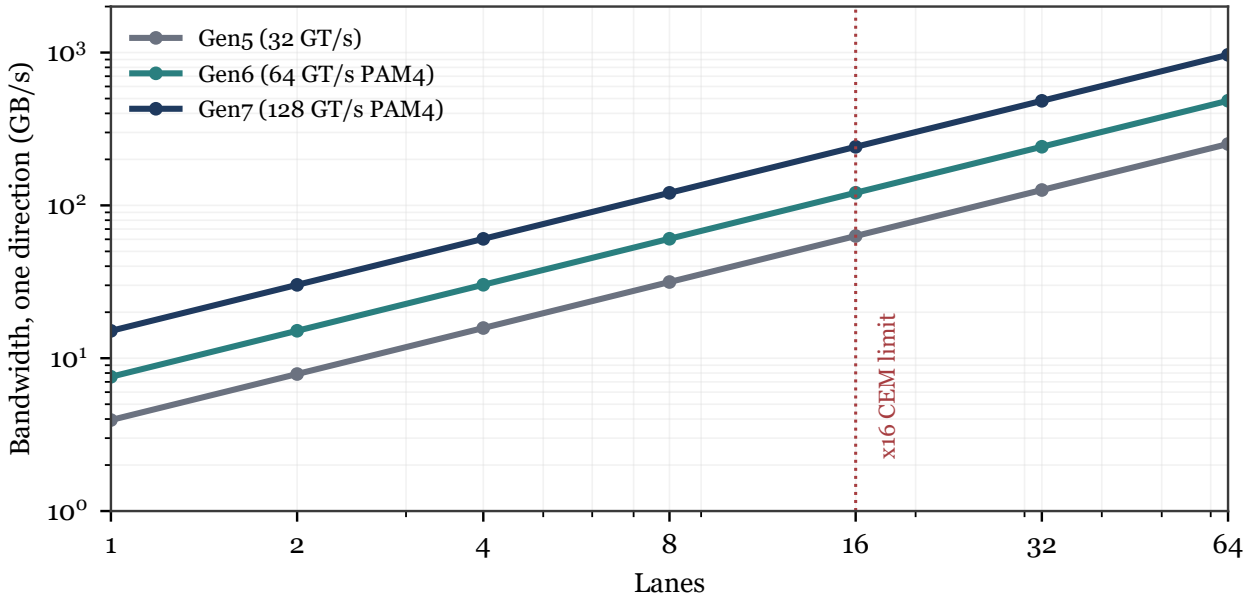


Figure 1: Bandwidth versus lane count for Gen5, Gen6, and Gen7. The x16 point is the top of the CEM connector; x32 and x64 are protocol-permitted but connector-limited. The line is linear in N; the cost is not.

Table 2: The width tax at x16, x32, and x64. Values are first-order estimates for a Gen6-class SerDes; connector pin counts for x32 and x64 are hypothetical.

Resource	x16	x32	x64	Scaling
Data pins	64	128	256	Linear
Ground pins	32	64	128	Linear
Package bumps	64	128	256	Linear
SerDes power (Gen6)	1.6 W	3.2 W	6.4 W	Linear
Retimers	0–2	2–4	4–8	Superlinear
Connector pins	164	328	656	Linear, but impractical
PCB layers	16	20–24	28–32	Superlinear
Local thermal density	1×	2×	4×	Linear in power, worse in airflow

four to eight. The total is modest against a 75 W slot, but it is not free: at x64, 10 W is 13% of the slot, and the card's compute budget shrinks.

3.5 Thermal

The thermal tax is local, not global. The connector and cable are often uncooled; their temperature rise is 10–20°C, and the cable jacket is rated at 85°C. The package hotspot is at the die edge, where the SerDes sits; the airflow shadow behind the connector is the worst place to put heat. At x64, the SerDes power is a small fraction of the card, but it is concentrated in the last centimeter, which is exactly where the thermal design is weakest [13].

3.6 Protocol

More lanes mean more deskew, more L2/L3 state, more arbitration, and more training. The per-lane overhead is small, but the tag space does not scale with lanes: a single function still has 256 tags, and a cache-line fetch is still 64 bytes [4]. More lanes raise the link ceiling; they do not raise the latency bound.

4. Fewer, Wider Lanes

4.1 Wider modulation

PAM4 doubles the bits per symbol at the same baud. It costs about 9.5 dB of SNR, which is why Gen6 pairs it with FEC and a flit-based framing [1]. PAM8 triples the bits per symbol and costs about 19 dB; at 32 GBd over a meter of

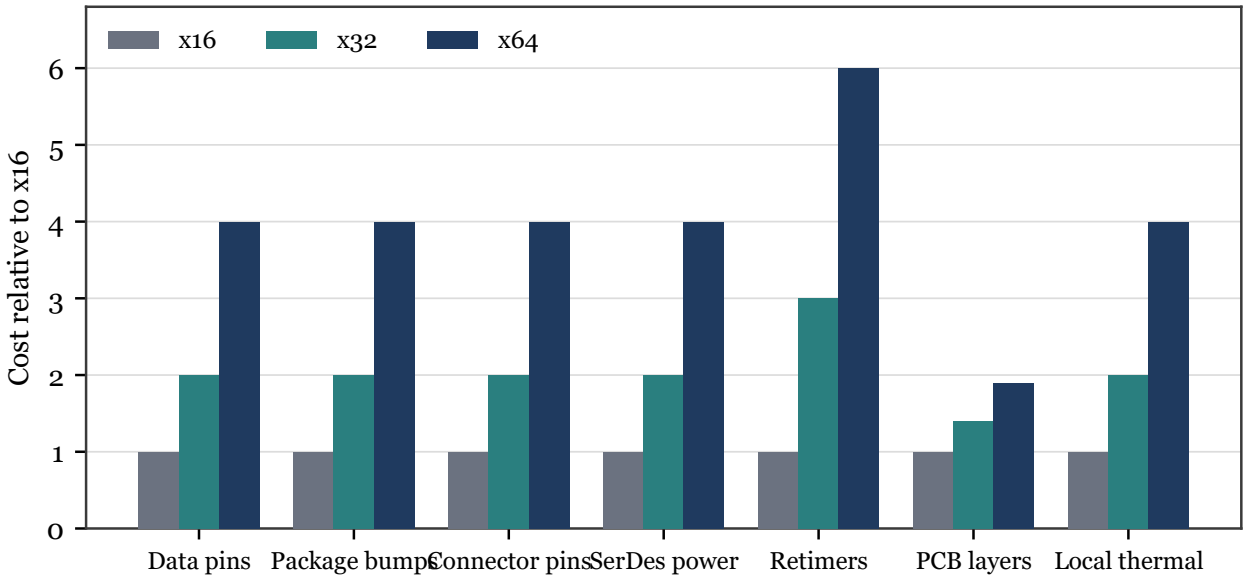


Figure 2: The width tax at x16, x32, and x64, normalized to x16. Pins and package bumps grow linearly; connector cost, retimer power, and local thermal density grow faster. The bars are first-order estimates.

Table 3: Routes for scaling a link. Each changes a different variable in $B = NRW\eta$ and pays a different tax.

Route	Bandwidth	Pins	Power	Reach	Verdict
More lanes	Linear	Linear	Linear	Same	Short reach only
Faster lanes	Linear	Same	Superlinear	Lower	Long reach with optics
PAM4	2×	Same	+	Lower	Gen6 default
PAM8	3×	Same	++	Much lower	Not yet
Wider parallel	Linear	Linear	Lower/bit	Short	Package and board
Merge 2:1	Same	Half	Same/higher	Lower	Connector niche
Halve width	Half	Half	Half	Same	Power and thermal
No lanes	Depends	Zero	Depends	Extreme	Package and network

FR4, the SNR is not there. The “wider lane” is really a deeper constellation, and it trades reach for rate.

4.2 Higher rate

Raising the baud from 32 to 64 to 128 GBd raises the loss. FR4 loses roughly 1 dB per inch at 16 GHz and 1.5 dB per inch at 32 GHz; the equalization budget is 20–30 dB, and beyond that the link needs a retimer or optics. The rate wall is the reach wall [12].

4.3 Fewer lanes at higher rate

An x8 link at 64 GT/s PAM4 carries the same bandwidth as an x16 link at 32 GT/s NRZ. It halves the pins and the connector, but it doubles the per-lane rate, halves the reach, and raises the per-lane power. This is the “less but wider” route: fewer, faster lanes. It is the right choice when the connector is the bottleneck and the channel is short.

4.4 Wider parallel lanes

The other meaning of “wider” is more wires per lane at a lower rate: a 16-bit-wide lane at 8 GT/s carries 128 Gb/s over 32 wires and a clock. It has no SerDes, no CDR, no equalization; it is a source-synchronous bus. It is the right choice at package and board reach, and it is what HBM and UCIe do [7, 6]. The lane disappears into a bus.

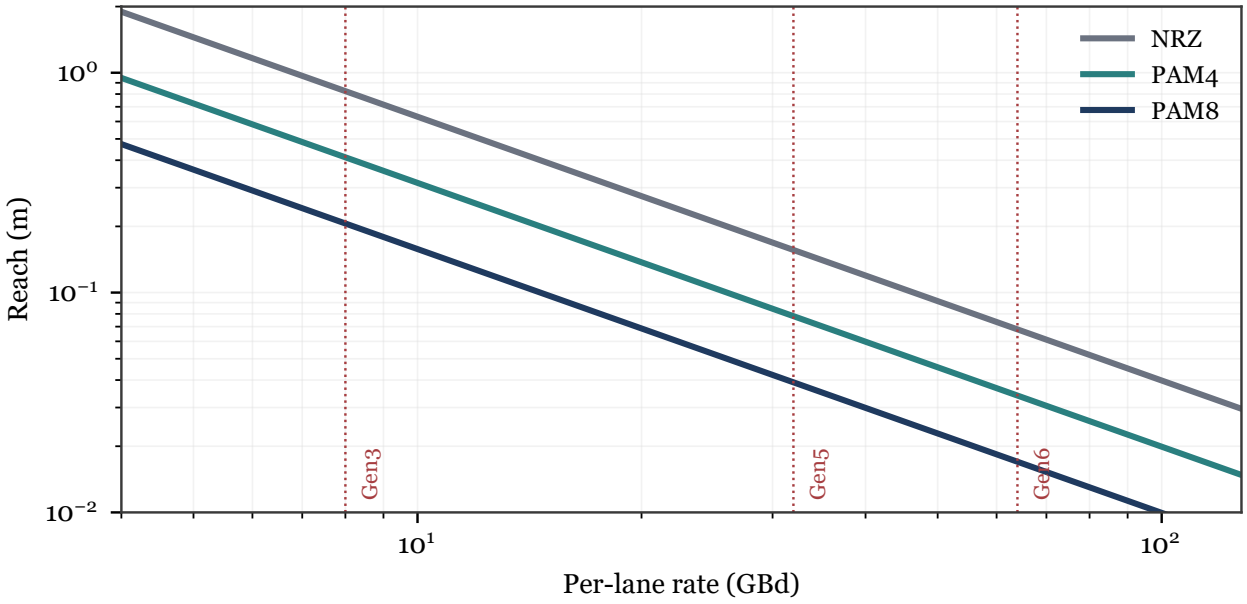


Figure 3: Reach versus per-lane rate for NRZ, PAM4, and PAM8. The curves are first-order loss budgets; the markers are PCIe generations. Higher modulation buys rate at the cost of reach, which is why Gen6 needs FEC and why Gen7 needs short channels.

5. Merging Lanes and Reducing Width

5.1 Aggregation

Lane aggregation combines physical lanes into one logical link. Reducing width from x16 to x8 halves the bandwidth and halves the pins and power. The latency is unchanged; the tag space is unchanged; the concurrency is unchanged. For latency-bound traffic, the effective bandwidth is the minimum of the link ceiling and the latency bound [4]; reducing width does not hurt until the link ceiling falls below the latency bound.

5.2 The effect

Figure 4 shows the effective bandwidth versus width for two workloads: 64-byte cache-line traffic with 256 tags at 1 μs, and 4 KB streaming traffic. The cache-line workload saturates at x4; the streaming workload scales with width. The lesson: width should be matched to the workload, not maximized.

5.3 Merging two lanes into one

Combining two lanes into one at twice the rate keeps the bandwidth and halves the pins, but doubles the per-lane rate, halves the reach, and raises the per-lane power. It is a connector and package optimization, not a bandwidth optimization.

5.4 Failover and dynamic width

PCIe already negotiates width. A broken lane reduces the link to x8 or x4; the system degrades gracefully. Dynamic width can also trade bandwidth for power: idle lanes can be turned off, and the link can widen when the workload needs it. The cost is training time and protocol complexity [1].

5.5 Recommendation

Match width to workload: x4–x8 for cache-line memory at microsecond latency, x16 for bandwidth-bound streaming, x32 only for short-reach backplanes where the connector is not the limit.

6. No Lanes at All

6.1 Optics

An optical link has no electrical lanes. It has wavelengths, fibers, and lanes in the WDM sense. The reach is 10 m to 10 km; the energy is 5–15 pJ/bit; the latency is set by the conversion, not the fiber. Co-packaged optics moves the conversion to the package, removing the electrical reach limit. The lane becomes a wavelength [14, 15].

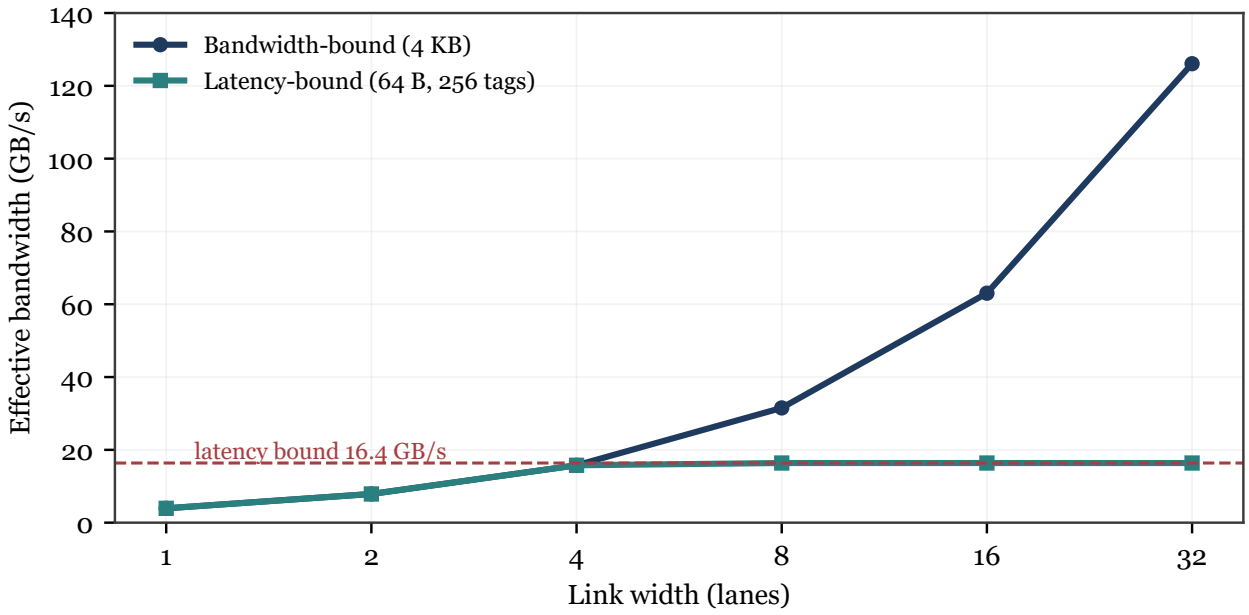


Figure 4: Effective bandwidth versus link width for a latency-bound cache-line workload and a bandwidth-bound streaming workload. Beyond x4, the cache-line workload gains nothing from more lanes; the streaming workload scales linearly.

Table 4: Lane-less alternatives. Reach, bandwidth, and latency are order-of-magnitude; the last column is the regime where each wins.

Technology	Reach	Bandwidth	Latency	Maturity	Regime
Optics	10 m–10 km	100 Gb/s–1.6 Tb/s	100–500 ns	Mature	Long reach
Wireless 60 GHz	1–10 m	1–10 Gb/s	100 ns	Niche	Rotation, reconfiguration
UCIe	0.5–2 mm	1–2 Tb/s	<10 ns	Emerging	Package
HBM	~5 mm	1.2 TB/s	<10 ns	Mature	Interposer
CXL over Ethernet	100 m–10 km	100–400 Gb/s	10–100 μs	Emerging	Shared network
Power-line data	1–10 m	1–100 Mb/s	10 μs	Research	Power pins

6.2 Wireless

60 GHz wireless carries 1–10 Gb/s over 1–10 m with no lanes and no connector. It is a niche for rotation and reconfiguration, not for memory.

6.3 Package-level

UCIe, HBM, and NVLink use no PCIe lanes. They use bumps, interposers, and short traces: 0.5–5 mm, 1–2 Tb/s, sub-10 ns. The lane becomes a bump. This is the fastest growing part of the interconnect space, and it is where the width tax is paid in silicon area rather than pins [6, 7, 8].

6.4 Network fabrics

CXL over Ethernet and RDMA use no dedicated lanes; they use a shared network. The reach is 100 m to 10 km; the latency is 10–100 μs; the bandwidth is 100–400 Gb/s per port. The lane becomes a packet [5].

6.5 Power-line communication

Data over the power delivery network uses no data lanes; it modulates the power pins. The bandwidth is Mbps, the reach is meters, and the noise is the VRM. It is a research route, not a memory route.

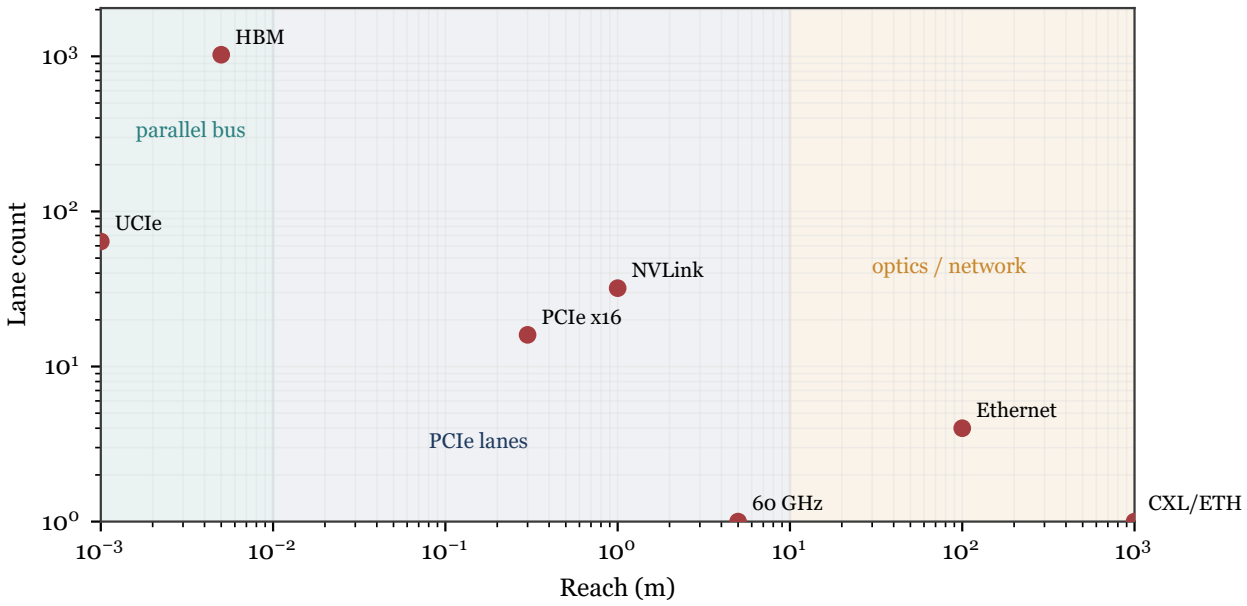


Figure 5: The lane frontier: lane count versus reach, with the regions where parallel buses, PCIe lanes, optics, and lane-less fabrics win. The markers are representative technologies, not products.

7. The Lane Frontier

7.1 The design space

The four variables are N, R, W, and L. The constraints are bandwidth B_0, reach L_0, power P_0, and pins N_0. The optimization is

$$\min_{N,R,W} P \quad \text{s.t.} \quad NRW\eta \geq B_0,\ L_{\max}(R,W) \geq L_0,\ N \leq N_0. \tag{6}$$

7.2 The frontier

The solution has a simple shape. At short reach, N^* is large and R^* is low: a parallel bus. At long reach, N^* is small and R^* is high: a serial lane with optics. In the middle, PCIe's lane count is a compromise. The width tax makes x32 and x64 unattractive except at the short-reach end.

7.3 The regions

Figure 5 maps the frontier. The parallel-bus region is below 1 cm; the PCIe region is 0.1–3 m; the optics region is above 10 m; the wireless and network region is the no-lane extreme. The boundaries are not sharp, but the trend is.

7.4 The lesson

The lane is not a law of nature; it is a packaging choice. The next PCIe generation should expose width, rate, and modulation as a triad, and it should let the platform choose the point on the frontier.

8. Recommendations

1. **Expose the triad.** Let a link negotiate width, rate, and modulation independently, so a platform can choose x8 at 64 GT/s, x4 at 128 GT/s, or x16 at 32 GT/s.

2. **Use modular connectors.** Two x8 connectors are cheaper and more flexible than one x16; four x8 connectors are the practical version of x32.

3. **Make width thermal-aware.** Turn off idle lanes; widen when the workload needs bandwidth; use width as a power knob.

4. **Use co-packaged optics for long reach.** Do not try to extend copper beyond the reach wall; move the conversion to the package.

5. **Standardize lane-less fabrics.** UCIe for the package, CXL over Ethernet for the network, and a clear boundary between them.

6. **Measure energy per bit per lane.** Bandwidth per pin is the wrong metric; energy per bit per meter is the right one.

9. Conclusion

More lanes is the obvious answer, and it is the wrong one at both ends of the reach spectrum. At short reach, a wide parallel bus is cheaper; at long reach, optics is cheaper; in the middle, PCIe's lane count is a compromise that the width tax makes increasingly expensive. The next decade of interconnect will be won by matching width to reach, not by maximizing lane count. The lane is a means, not an end.

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